Key concepts — Session B: backup, restore, recovery plan

A short study guide in plain English, with the French terms in brackets and examples from your own lab. Charles Maduba — AIS training, GRETA Essonne, October 2026.

1. Backup, replication and HA are not the same thing

	Backup (sauvegarde)	**Replication (réplication)**	**HA (haute disponibilité)**
What it is	A dated copy you can restore later	A copy kept up to date continuously or every few minutes	Automatic restart of a VM on another node
Protects against	Deletion, ransomware, corruption, site loss (if off-site)	Loss of a node or disk	Failure of a node
Does NOT protect against	Data written since the last backup	Deletion or ransomware (the damage is copied too)	Deletion, ransomware, loss of the storage
Typical RPO	Time since the last backup (e.g. 24 h)	The replication interval (e.g. 15 min)	≈ 0 with shared storage

In your lab: *HA restarted VM-Test in 2 min 17 s in session A, but it could not help if someone deleted the VM: only the 21:00 backup could.*

2. Proxmox backups: vzdump

- **vzdump** — Proxmox's backup tool. The web interface "Backup now" button runs it for you.
- **Backup modes** — snapshot: the VM keeps running (no interruption); suspend: the VM is paused briefly; stop: the VM is shut down for the most consistent copy.
- **Compression (ZSTD)** — makes the archive smaller and is fast. An empty 2 GB disk became a 72 KB file.
- **Archive name** — vzdump-qemu-<VM id>-<date_time>.vma.zst, plus a .log file. Stored in /var/lib/vz/dump on the "local" storage.
- **Where the backup lands** — on the storage you choose, on the node running the VM. If that node is destroyed, its local backups go with it.

In your lab: *vzdump 100 --storage local --mode snapshot --compress zstd took 10 seconds, and VM-Test stayed "running".*

3. Restoring: qmrestore

- **Test restore (restauration de test)** — the only real proof that a backup works. An untested backup proves nothing.
- **New ID** — restore under another number (900) so you never overwrite the running VM or confuse HA.
- **Unique** — gives the restored VM a new MAC address and identifiers, avoiding network conflicts with the original.
- **Never use "Restore" on the running VM's Backup tab** — it overwrites the existing VM.
- **Measuring it** — put "time" in front of a command; the "real" line is the elapsed time.

In your lab: *qmrestore … 900 --storage local-lvm --unique 1 took 3.6 s. A full 50 GB production disk would take minutes, not seconds.*

4. Scheduled backups and retention

- **Backup job (tâche de sauvegarde)** — a backup that runs by itself on a schedule (e.g. every day at 21:00). Stored in /etc/pve/jobs.cfg.
- **Retention / prune** — how many backups to keep. keep-last=7 keeps the 7 newest and deletes older ones automatically, so the disk never fills.
- **GFS idea** — real setups often mix keep-daily, keep-weekly and keep-monthly to go further back in time with few files.
- **Monitoring** — a scheduled backup must be checked: alert by email or Zabbix if it fails.

5. The 3-2-1 rule

- **3 copies** — the production data plus 2 backups.
- **2 different media** — e.g. server disks plus a separate backup server or NAS.
- **1 off-site copy** — survives a fire or theft at the main site.
- **Immutability (immuabilité)** — a copy that cannot be modified or deleted for a set time, the best defence against ransomware.
- **Proxmox Backup Server (PBS)** — Proxmox's dedicated backup server: deduplication, verification, encryption, off-site sync. Chosen in your requirements spec.

In your lab: *3 copies, but all on one laptop disk and no off-site copy: the lab does not follow 3-2-1.*

6. RTO and RPO

	RTO — Recovery Time Objective	**RPO — Recovery Point Objective**
Question	How long can the service be down?	How much data can we lose?
Measured as	Time from failure to service back	Time between the last usable copy and the incident
Improved by	HA, fast restores, written procedures	More frequent backups, replication
Your lab	2 min 17 s (node failure, HA); 3.6 s restore (empty disk)	≈ 0 (node failure); up to 24 h (restore from the daily backup)

Rule of thumb: the backup frequency sets the RPO. Daily at 21:00 means up to 24 h of data lost; a VM deleted at 18:00 loses about 21 h.

7. PCA and PRA

	PCA — Plan de Continuité d'Activité (Business Continuity Plan)	**PRA — Plan de Reprise d'Activité (Disaster Recovery Plan)**
Goal	Stay in service during the incident	Come back into service after a disaster
Tools	Cluster, HA, redundancy	Backups, off-site copies, restore procedure
Example	A node fails, the VM restarts elsewhere	Ransomware or fire: restore from a clean backup

A PRA document has 6 sections: 1. scope (which services and VMs), 2. disaster scenarios with RTO/RPO, 3. step-by-step restore procedure, 4. roles (who decides, who restores, who communicates), 5. targets compared with measurements, 6. testing and updating. It must be usable by someone who did not build the system.

8. Lab troubleshooting from this session

Symptom	**Cause**	**Fix**
VirtualBox "Guru Meditation", error HostMemoryLow	Not enough free RAM on the host PC to lock memory for the VMs	Close other apps (browsers, chat apps); start nodes one at a time
pvecm status says Quorate: Yes, but ha-manager says "No quorum" with old timestamps	pmxcfs (/etc/pve) stuck after boot	systemctl restart pve-cluster, then check ha-manager status
"qm config 100" says the file does not exist	qm only sees VMs on the node where you type it	Run it on the node that hosts the VM
Start time of a command scrolled off the screen	Long output	Prefix the command with "time"